AI Deep Drawability

July 2026 Project Update

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1 Updates

My first change this week was updating my model to include the label output as a final layer after the sheet classification. Previously I had attempted a model that predicts sheet and label separately and a model that predicts sheet and uses a stored dictionary to convert those predictions to labels. This new model allows back propagation to update the weights for not just my sheet outputs but also the good bad label which will help allow the model to generalize. This lead to very minor performance improvements over the two previous attempts.

The next step was hyperparameter tuning for this new model. Until this point I had been using the best settings found from the MLP model assuming they would be similarly effective on this one. Now I wanted to test some different configurations to be sure. I used Baysian optimization and found that a two hidden layer approach with 128 and 64 neurons performed better (although only by a small margin so this may be just be overfitting). Additionally, I found that adding an intermediate 16 neuron layer between the sheet layer and final label output was beneficial. This is likely because it allows the model to better learn possible non-linear relationships between those two layers.

Figure 1 shows the updated model architecture visually. This is the architecture for a single holdout model and we train 15 of these; one for each possible holdout combination.

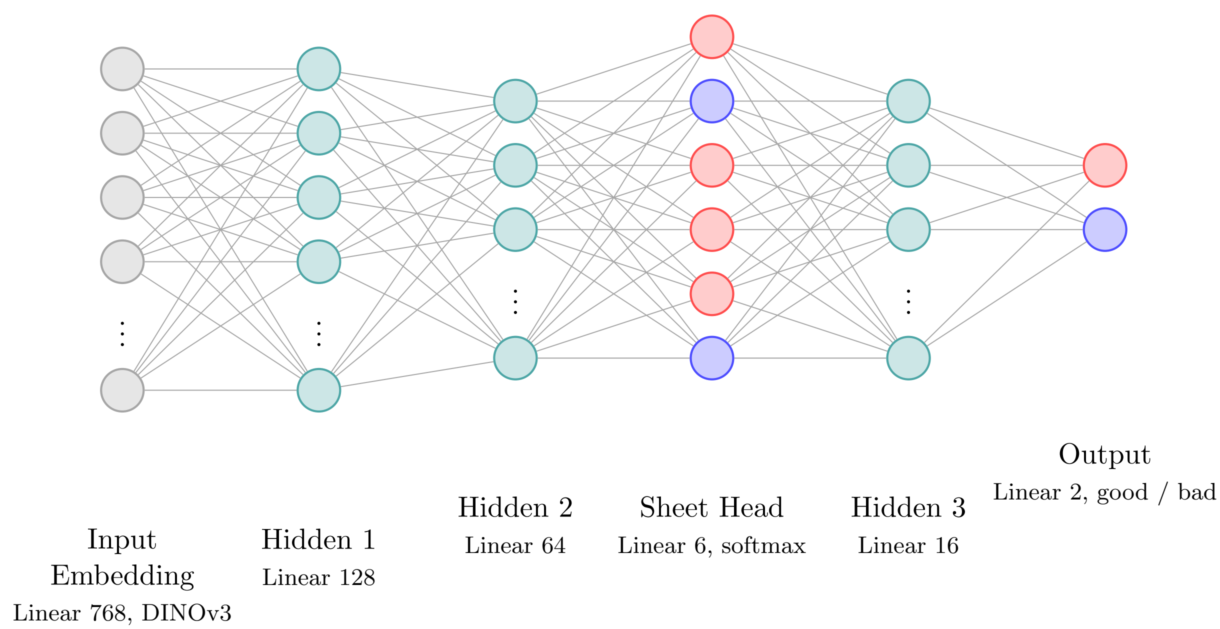

Figure 1: Sheet and label classifier neural network architecture

We start with our stored 768-dimensional embeddings, extracted by feeding raw images through Meta's pretrained DINOv3 model. These pass through 2 hidden layers (128 $\rightarrow$ 64), which feed into the sheet head. This is a linear layer producing one logit per known sheet. Softmax converts those logits into probabilities that sum to 1 across all sheets. We then feed those probabilities through a final hidden layer, which feeds into the output layer of 2 neurons giving our final good/bad classification. We can see the sheet head always has 2 good sheets and 4 bad as we hold out 1 good and 1 bad sheet from the total 3 and 5 we have. This means each model sees two times as many bad samples as good which is why we consistently see better accuracy on the bad sheets even when weighting our loss term proportionally.

After these changes we are looking at an accuracy of 0.711 on the bad samples and 0.541 on the good samples. This is our best result yet with a combined accuracy of 0.626. While this is an increase from our previous best of 0.612 it is still only a small shift. One note I want to make here, up until now I've been reporting the combined accuracy by weighing good and bad sheets evenly but since we have a skewed distribution this isn't an accurate view of overall performance. The true average accuracy here across all holdouts is **0.635** with a standard deviation of **0.164**. Our best holdout accuracy sits around 0.82 while our lowest is under 0.3 hence the high standard deviation.

Figure 2 below shows the spread of all holdout accuracies. Each holdout pair point is plotted with jitter to make it more readable. The left half is colored based on the held out good sheet, while the right is colored based on the held out bad sheet.

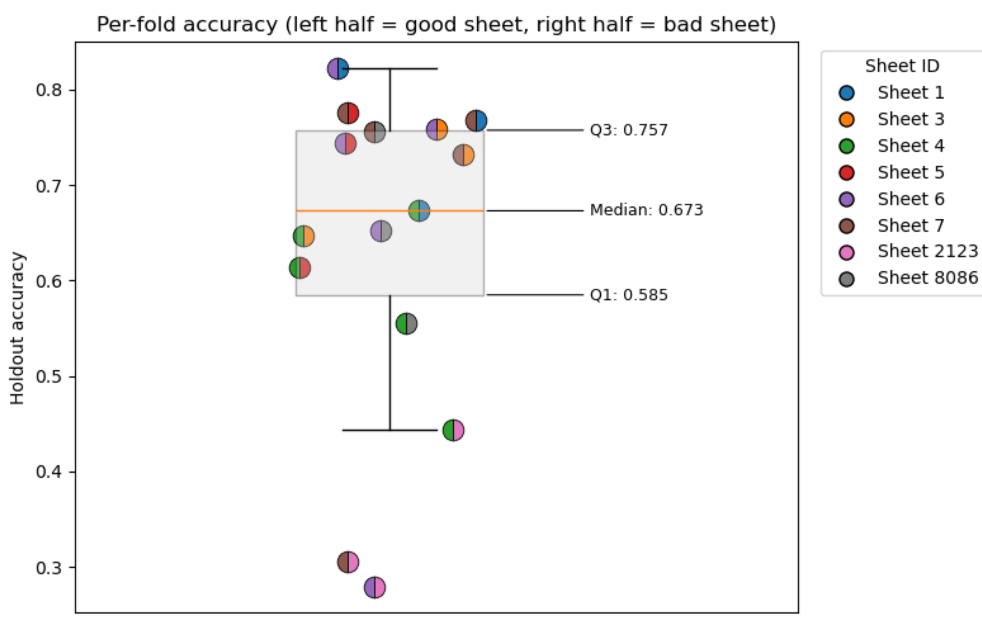

Figure 2: Holdout accuracy spread

We can see that for 7 of the holdouts we are over 70% prediction accuracy and for a further 5 we are above 50% and only 3 sit below 50%. This overall strong performacne is reflected in our median accuracy of 0.673 and the IQR sitting well above 0.50. Looking into our worst performers, all 3 of them are when sheet 2123 is held out. If we were to drop this sheet our overall accuracy goes to **0.708** and our standard deviation drops to just **0.075**. But sheet 2123 is genuine data and shows how different samples can have different underlying structures that our model can't immediately recognize. Dropping it would be cherry-picking data to make our model appear better than it is. This is why a large and diverse dataset is so important. We want our model to have seen many different sheets so it can overcome sample to sample variance and focus on the characteristics that make a sheet good or bad.

To dive a bit deeper into our current performance, Figure 3 shows the holdout accuracy per sheet combination for this new model. We can see a handful of good and bad sheets that stand out as consistent good performers. From the good sheets, 6 and 7 are consistently in the high 0.70 range. And for the bad 1,3, 5, and 8086 perform similarly well. This tells us that sheets 6 and 7 are likely similar so regardless of which is held out, the other provides the model with enough information to get decent accuracy on the unseen sheet. I believe the same is true for sheets 1,3,5, and 8086. Sheets 4 and 2123, on the other hand seem to have some more unique underlying characteristics. So when our model is trained without any samples from these images it performs more poorly once tested on them. For sheet 2123 specifically, our model still predicts good 100% of the time which is why the accuracy is so low.

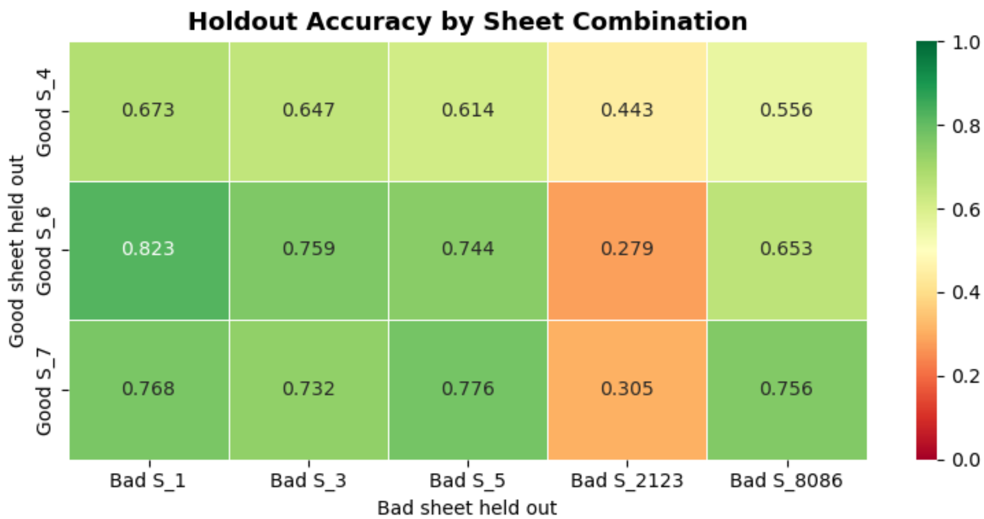

Figure 3: Per-holdout accuracy

Another way we can look at this is through a confusion matrix across all folds in Figure 4. Each holdout combo is tested on a model that has never seen that actual sheet and doesn't even have a logit associated with the true sheet in it. So our diagonal here will be all 0s as it is impossible for our model to correctly classify a sheet id it's never seen. What we are looking for is what sheets seem to be most similar. And specifically, we want to find cases where good sheets are most similar to bad and vice versa. To make this simpler to see, red cells here represent cases where a sheet is found to be most similar to one of the opposite class while blue represent cases where it is most similar to a sheet of the same class. The shade reflects the fraction of that sheet's held-out images matched to each predicted sheet, where darker means a larger share of that sheet's images landed on that match.

In Figure 4, we can see that sheets 6 and 7 are truly similar with both being classified as the other the majority of the time and both are good. Sheets 1, 3, and 5 also appear to be relatively similar and all are bad. Sheet 8086 is similar to 3 and 5 but less similar to 1. Sheet 2123 immediately stands out as it is most similar to sheet 4. This is problematic because sheet 4 is a good sheet while 2123 is bad. This explains the low accuracy we see when holding out sheet 2123. Sheet 4 also stands out, being most similar to 8086 and 3 which are both bad sheets despite being good itself. This confusion matrix confirms our earlier hypotheses from the accuracy visual and gives more context on which sheets are being mislabeled as others. Overall, we can see that our hope that new good sheets would be more similar to known good sheets and vice versa seems to hold with 6/8 sheets

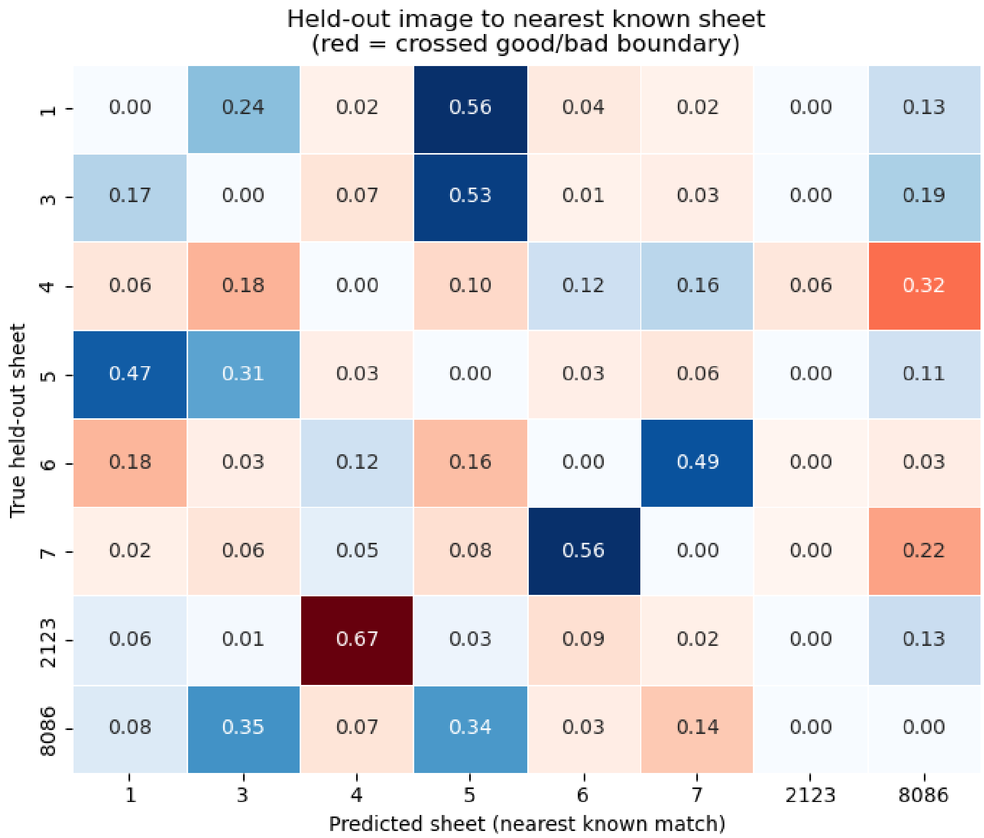

Figure 4: Cross-holdout confusion matrix

having the highest probability for a sheet of the same class. However, the 2/8 that are misclassified as the opposite class are key shortcomings in our current model and show that we still need further refinement.

2 Conclusion and Next Steps

Overall, with these architectural improvements and tuning steps we are seeing small performance improvements each week. At an average accuracy of 0.635 and median of 0.673 our model is clearly learning meaningful information about these images and, on the whole, performing well considering the quantity of data currently available. Looking at the data more closely it's clear that we still have a couple sheets that are lowering the aggregate performance considerably. Identifying which sheets are most often misclassified as the opposite class is extremely valuable and this data will be used in the coming weeks to hopefully improve the performance on these worst performing sheets. Sheets 4 and especially 2123 clearly have a unique structure within their class and warrant further exploration. One future avenue that will hopefully help address this is additional data from unique sheets but for now I will attempt manual tuning to overcome these issues.